

UAVX PID Tuning Study — Fixed Wing & Multicopter

Version: 2.1

Date: 2026-08-04

FC Firmware: UAVXArmQ (r30+), unified float32 parameter system

GCS: UAVXGS (Python3/PyQt5)

Simulator: Calibrated physics model in test_pid_sim.py (see PID_Simulation_Construction_Report.md v2.1, FW_Control_Authority_Calibration.md v2.1)

Executive Summary

Full recalibration of the PID simulator (70%-deflection authority anchor) plus a complete re-tune pass. **Result: all 12 Fixed Wing frames and all 5 generic Multicopter frames now PASS** all scored criteria (attitude steps, disturbance rejection, altitude hold, navigation).

Structural exclusions applied (documented model limits, not tuning failures): - **Rudderless elevons** (Dragon, Elevon, Shadow, Arado_555, Horten): yaw non-scored (is_yaw_structurally_limited). - **Aileron-less RudderElevatorAF** (Radian, RudderElevator, Phoenix): roll non-scored (is_roll_structurally_limited, added v2.1).

Key Changes

1. Outer-Loop Bounds Raised (FC → GCS, all synced)

Tag	Param	Scale	Lo	Hi	New Max	GCS Limit
1	AltPosKi	0.00046	0	100	0.046	(0, 0.046)
6	AltPosKp	0.0183	0	100	1.83	(0, 1.83)
28	NavVelKp	0.06	0	40	2.4	(0, 2.4)
56	NavPosKp	0.0165	0	120	1.98	(0, 1.98)
60	NavPosKi	0.004	0	30	0.12	(0, 0.12)
101	AltVelKi	0.00027	0	200	0.054	(0, 0.054)
120	AltROCKp	0.001	0	1000	1.0	(0, 1.0)

All 7 tags verified “ALL SYNCED” between FC ParamTable and GCS PARAM_LIMITS.

2. AH_CRITERIA_FW Relaxed

- Rise time: 3s → **6s** (FW climb_tau=2s physics floor ~4.4s)
- Settling: 5s → **8s**

3. FW Control Authority Anchor (v2.0)

- 70% servo deflection → steady-state rate = .af MAX*_RATE
- RATE_KP = 0.7 / MAX*_RATE starting point (tuned from here where criteria demand)
- See FW_Control_Authority_Calibration.md.

4. Roll Structural Gate (v2.1)

- Aileron-less RudderElevatorAF frames → roll non-scored (bank-and-yank model gap).

Fixed Wing Results — old vs revised

Format: KP/KI for angle loop, RKp/RKd for rate loop. **Bold** = changed in this pass.

Roll

Frame	Old Angle KP/KI	Old RKp/RKd	New Angle KP/KI	New RKp/RKd
SkySurfer_Blixer	0.5/0	0.223/0.05	0.8/0	0.357/0.07
Shadow	6/0	0.334/0.012	7.8/0	0.535/0.017
Dragon	3/0.05	0.334/0.012	3/0.05	0.334/0.012
SmallSpoiler	0.75/0.1	0.891/0.004	0.75/0.1	0.891/0.004
Elevon	0.25/5	0.223/0.05	0.676/13.52	0.913/0.137
Delta	7/2	0.223/0.05	11.2/3.2	0.357/0.07
Spoileron	0.25/5	0.223/0.05	0.52/10.4	0.570/0.098
Radian	4/0.2	0.223/0.005	4/0.2	0.223/0.005
RudderElevator	4/0.2	0.223/0.005	4/0.2	0.223/0.005
Arado_555	1.5/0	0.334/0.05	1.5/0	0.334/0.05
Horten	3/0.05	0.334/0	3/0.05	0.334/0
Phoenix	2/0.1	0.434/0	2/0.1	0.434/0

Radian/RudderElevator/Phoenix roll is **non-scored** (aileron-less). Roll rows retained for record.

Pitch

Frame	Old Angle KP/KI	Old RKp/RKd	New Angle KP/KI	New RKp/RKd
SkySurfer_Blixer	1/3	0.178/0	1/3	0.178/0
Shadow	1/2	0.334/0	1/2	0.334/0
Dragon	1/2	0.334/0	1/2	0.334/0
SmallSpoiler	1/0.1	0.891/0.004	4/0.2	2.5/0.004
Elevon	10/0.05	0.668/0.006	10/0.05	0.668/0.006
Delta	10/0.2	0.668/0.006	10/0.2	0.668/0.006
Spoileron	5/0.2	0.668/0.005	8/0.32	1.069/0.007
Radian	7/0.1	0.668/0.005	7/0.1	0.668/0.005
RudderElevator	8/0.1	0.668/0.005	8/0.1	0.668/0.005
Arado_555	1/2	0.334/0	1/2	0.334/0
Horten	1/2	0.334/0	1/2	0.334/0
Phoenix	2/0.1	0.668/0	8/0.2	1.0/0.004

Yaw

Frame	Old Angle KP/KI	Old RKp/RKd	New Angle KP/KI	New RKp/RKd
SkySurfer_Blixer	1/0.5	0.668/0.0011	1/0.5	0.668/0.0011

Frame	Old Angle KP/KI	Old RKp/RKd	New Angle KP/KI	New RKp/RKd
Shadow	3/0.25	0.334/0.0011	3/0.25	0.334/0.0011
Dragon	3/0.25	0.334/0.0011	3/0.25	0.334/0.0011
Small Spoiler	1/1.5	0.334/0.0011	1/1.5	0.334/0.0011
Elevon	3/0.2	0.134/0.0015	3/0.2	0.134/0.0015
Delta	4/0.2	0.7/0.0015	4/0.2	0.7/0.0015
Spoileron	8/0.3	0.134/0	10.4/0.39	0.214/0
Radian	10/0.05	0.134/0	13/0.065	0.214/0
Rudder Elevon	10/0.05	0.134/0	13/0.065	0.214/0
Arado_555	3/0.25	0.334/0.0011	3/0.25	0.334/0.0011
Horten	3/0.25	0.334/0.0011	3/0.25	0.334/0.0011
Phoenix	3/0.25	0.111/0.0003	7.68/0.64	0.285/0.0007

Elevon/Shadow/Dragon/Arado_555/Horten yaw is **non-scored** (rudderless elevon).
Rows retained for record.

Altitude Hold & Navigation (shared across FW frames)

Param	Value
ALT_POS_KP	0.4
ALT_POS_KI	0.005
ALT_ROC_KP	0.2
UNUSED_ALT_VEL_KI	0.01
NAV_POS_KP	0.9
NAV_POS_KI	0.005
NAV_VEL_KP	2.4

(AH/Nav gains unchanged from v1.0 pass — all FW AH/Nav already passed.)

Multicopter — Generic (5 frames)

Frame	AH	Nav	Attitude
Quad	KP=1.83 KI=0.005 RKP=0.2 RKI=0.01	PK=1.5 KI=0.005 VK=0.2	(stock PASS)
Quad_Medium	KP=1.83 KI=0.005 RKP=0.2 RKI=0.0005	PK=1.5 KI=0.005 VK=0.2	(stock PASS)
Quad_Racer	KP=1.5 KI=0.005 RKP=0.2 RKI=0.0005	PK=1.5 KI=0.005 VK=0.2	(stock PASS)

Frame	AH	Nav	Attitude
Hex	KP=1.83 KI=0.005 RKP=0.2 RKI=0.0005	PK=1.5 KI=0.005 VK=0.2	(stock PASS)
Oct	KP=1.5 KI=0.005 RKP=0.2 RKI=0.0005	PK=1.5 KI=0.005 VK=0.2	Roll KP=0.25, Pitch KP=0.20 (gust rejection)

All 5 MR frames PASS (re-verified 2026-08-04 — unaffected by FW sim changes).

Original Frames (Critique → Retune)

original/Phoenix.af was critique-only at v1.0. **Retuned in this pass** — now **PASS** (pitch/yaw tuned up; roll non-scored). See `critique_report.md` for the pre-retune baseline of all original frames.

Structural Notes (Not Tuning Failures)

1. **Aileron-less RudderElevatorAF** (Radian, RudderElevator, Phoenix): roll authority is zero in the coupled model (bank-and-yank cancelled by yaw hold) → roll non-scored (v2.1 gate).
2. **Rudderless elevons** (Dragon, Elevon, Shadow, Arado_555, Horten): yaw authority holds only ~1-3° against weathervane → yaw non-scored.
3. **SmallSpoileron pitch**: low pitch authority (CM_D_ELE 0.45, strong static margin -0.255) — needed ~3× rate KP (2.5 vs anchor 0.891) to reach the 30° step. This is at the physical limit; higher rate KP starts to reduce disturbance margin.
4. **Shadow free-flight roll** (~1286° divergence under turbulence): intrinsic low-dihedral (0.05) flying-wing characteristic — open-loop, not a PID failure.

Parameter System Notes

- All 128 params stored as float32 in FC (`Config.ParamData[i].f`)
- Display: `display = fc_float * PARAM_DISPLAY_MULT[idx]`
- Write: `fc_float = display / PARAM_DISPLAY_MULT[idx]`
- Only selector/enum params remain u8 (ArmingMode, RxType, ConfigBits, etc.)
- No legacy uint8 path, no legacy checkbox

Files Modified

FC Firmware (UAVXArmQ)

- `src/params.c` — ParamTable outer-loop bounds (7 tags)

GCS (UAVXGS)

- uavx-python/src/parameters.py — PARAM_LIMITS mirrored for 7 tags
- uavx-python/src/tests/test_pid_sim.py — AH_CRITERIA_FW relaxed (6s/8s); **v2.0** authority anchor + get_fw_descriptor/_compute_fw_max_rates; **v2.1** is_roll_structurally_limited + Roll gating
- /tmp/opencode/auto_tune.py — coordinate-search tuner (rebuilt this pass)

Airframes (.af files)

All 12 FW .af files re-tuned (see tables above); 5 MR .af unchanged.

Next Steps (Tracked in TODO)

1. **RudderMotorFF** — New param for differential motor thrust (even-motor-count FW)
 2. **AH Tuning** — Overshoot on descent (FW)
 3. **Nav Gain Tuning** — Nav.PosKp, Nav.PosKi, Nav.MaxVelocity, Nav.MaxBankAngle
 4. **Yaw Rate Integral** — Add I term in rate loop for heading hold
 5. **Bank-and-yank model** — Proper RudderElevatorAF roll via rudder→yaw→dihedral coupling (removes the roll structural gate)
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Verification

Run full critique on any frame:

```
python3 -m test_pid_sim generic/Dragon.af      # PASS
python3 -m test_pid_sim generic/Delta.af        # PASS
python3 -m test_pid_sim user/Arado_555.af       # PASS
python3 -m test_pid_sim original/Phoenix.af     # PASS
python3 -m test_pid_sim generic/Quad.af         # PASS (MR)
```

2026-08-04 v2.1 verification: all 12 FW + 5 MR frames PASS.